AP Tech Purchase Process

Analysis & AI Solution Proposal

Process Restructuring Through AI Technologies

Technical Team Document

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CONFIDENTIAL

1. Executive Summary

This document presents a comprehensive analysis of AP Tech's current purchase communication process and proposes an AI-powered solution to address critical inefficiencies. The current process relies heavily on the Purchasing Department as a manual relay between Engineers and Suppliers, resulting in excessive email chains (30+ emails per purchase), extended cycle times (approximately 2 weeks from request to confirmation), and an inability to process multiple purchase requests in parallel.

The proposed solution introduces an AI Purchasing Agent that acts as the central communication hub, automating supplier outreach, enabling parallel processing of purchase orders, and providing structured conversation logging. The implementation follows a phased approach, with an MVP deliverable in 6 weeks that addresses the most critical pain points: eliminating the manual email relay, querying the internal supplier database, and producing markdown-formatted conversation logs.

2. Current Process Overview

2.1 Process Flow Diagram

The diagram below illustrates the current end-to-end purchase process at AP Tech, organized by the three key actors (Engineer, Purchasing Department, and AP Tech Suppliers) across five sequential phases.

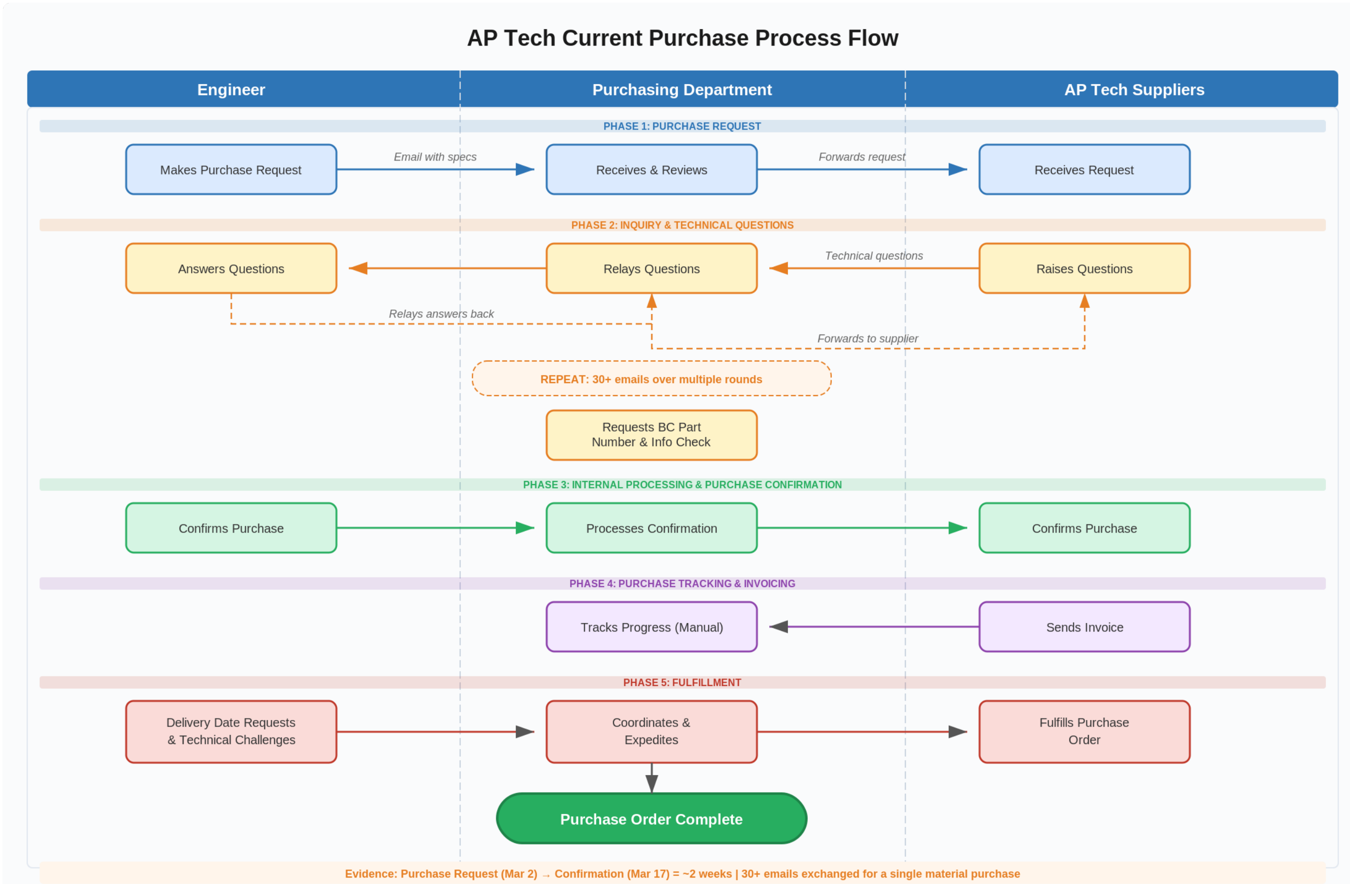

Figure 1: *Current AP Tech Purchase Process Flow Diagram*

2.2 Process Steps Summary

The current purchase process follows five distinct phases:

Phase 1: Purchase Request

The Engineer identifies a material need and submits a purchase request via email to the Purchasing Department. The Purchasing Department reviews the request and forwards it to one or more AP Tech Suppliers. At this stage, the process is strictly sequential — no parallel outreach to multiple suppliers occurs.

Phase 2: Inquiry & Technical Questions

This is the most time-consuming phase. Suppliers raise technical questions about specifications, compatibility, or part numbers. These questions are relayed through the Purchasing Department to the Engineer, who responds. The answers then travel back through Purchasing to the Supplier. This relay cycle repeats multiple times, with each round requiring at least two intermediary emails (Supplier → Purchasing → Engineer, then Engineer → Purchasing → Supplier). The Purchasing Department also handles internal tasks such as requesting BC part numbers and performing information checks.

Phase 3: Internal Processing & Purchase Confirmation

Once all technical questions are resolved, the Engineer confirms the purchase. The Purchasing Department processes this confirmation internally and communicates it to the Supplier, who acknowledges and confirms the order.

Phase 4: Purchase Tracking & Invoicing

After confirmation, the Supplier issues an invoice. The Purchasing Department manually tracks the progress of the order, updating internal records and following up as needed. This tracking is done through email and spreadsheets, with no automated system in place.

Phase 5: Fulfillment

During fulfillment, the Engineer may request delivery date updates or expedites, and technical challenges may arise that require discussion. All of these communications flow through the Purchasing Department. The process concludes when the purchase order is fully delivered.

3. Inefficiency Analysis

3.1 Identified Inefficiencies

The analysis identifies four primary inefficiencies in the current process:

#	Inefficiency	Root Cause	Impact
1	Manual information relay between Engineers and Suppliers	Purchasing Dept acts as a pass-through for every message, adding latency to each communication round	Each Q&A cycle requires 4 emails instead of 2; delays compound over multiple rounds
2	Inability to parallel process purchase requests	Sequential workflow — each purchase is handled one at a time by the same team	Purchasing becomes a bottleneck; engineers wait in queue for their requests to be processed
3	Slow supplier outreach	Manual identification and contact of potential suppliers; limited to known contacts	Missed opportunities for better pricing, lead times, or quality from unknown suppliers
4	Manual and error-prone order tracking	Tracking done via email and spreadsheets with no automated system	Orders can fall through the cracks; no proactive alerts for delays or issues

Table 1: *Inefficiency Summary Matrix*

3.2 Supporting Evidence

Two key data points substantiate these inefficiencies:

Evidence 1: Excessive Email Volume

A single material purchase required **30+ emails** to complete. In an efficient process with direct engineer-supplier communication, the same purchase would require approximately 8–12 emails. The 2.5–3x overhead is directly attributable to the relay function of the Purchasing Department, where every message must pass through an intermediary.

Evidence 2: Extended Cycle Time

The purchase request was initiated on **March 2** and the purchase confirmation was only received on **March 17** — a span of **approximately 2 weeks**. For a standard material purchase, industry benchmarks suggest a 3–5 business day cycle. The extended timeline is caused by compounding relay delays, sequential (not parallel) supplier engagement, and lack of automated follow-ups when responses are delayed.

3.3 Cost of Inaction

If these inefficiencies are not addressed, AP Tech faces compounding costs: purchasing staff spend the majority of their time on low-value relay work instead of strategic sourcing, engineers experience project delays waiting for materials, the organization misses better supplier options due to limited outreach capacity, and manual tracking increases the risk of missed deliveries or duplicate orders.

4. Proposed AI Solution

4.1 Solution Overview

The proposed solution is an AI Purchasing Agent that serves as the intelligent central hub for all purchase communications. Rather than replacing the Purchasing Department, the AI augments their capabilities by automating the repetitive relay and tracking work, freeing the team to focus on supplier relationship management, strategic sourcing, and exception handling.

4.2 System Architecture

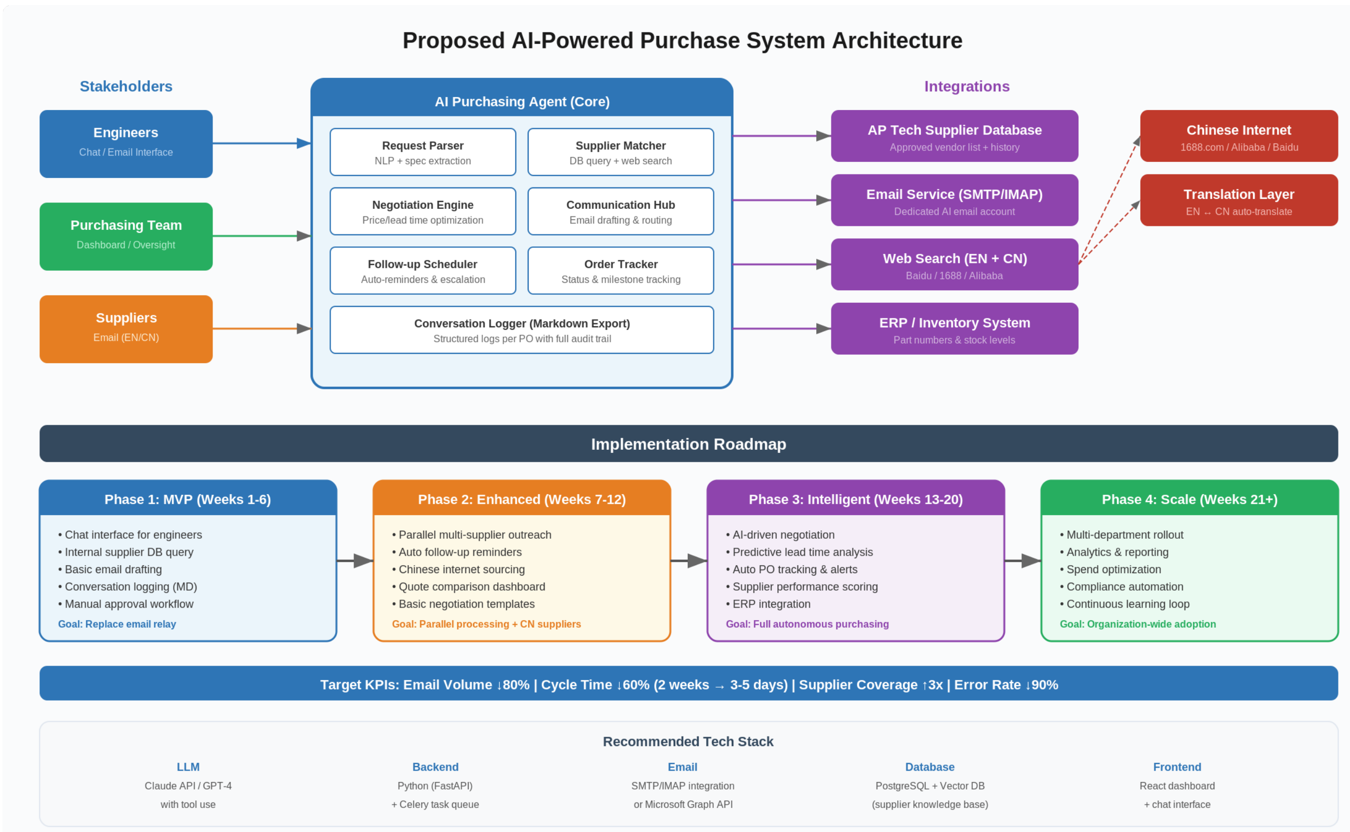

Figure 2: *Proposed AI-Powered Purchase System Architecture*

Core Modules

The AI Purchasing Agent consists of seven interconnected modules:

- **Request Parser:** Uses NLP to extract material specifications, quantities, urgency levels, and constraints from engineer requests submitted via chat or email.
- **Supplier Matcher:** Queries AP Tech's internal approved supplier database and performs web searches (including Chinese internet sources like 1688.com and Alibaba) to identify the best-fit suppliers based on the parsed requirements.

- **Communication Hub:** Drafts and sends emails to suppliers in the appropriate language (English or Chinese), manages responses, and routes relevant information to engineers and the purchasing team.
- **Negotiation Engine:** Compares quotes across suppliers on price, lead time, and quality metrics. In later phases, conducts autonomous negotiation within pre-approved parameters.
- **Follow-up Scheduler:** Monitors response times and automatically sends follow-up reminders to suppliers or engineers when responses are overdue. Escalates to the purchasing team if thresholds are exceeded.
- **Order Tracker:** Maintains real-time status of all purchase orders, tracks milestones (order placed, shipped, delivered), and generates alerts for delays.
- **Conversation Logger:** Records all interactions in structured markdown format, organized by purchase order. This provides a full audit trail and satisfies the requirement for retained, searchable conversation history.

4.3 Recommended Tech Stack

Layer	Technology	Rationale
LLM	Claude API (Anthropic) with tool use	Strong reasoning, native tool-use support for DB queries and email operations, excellent multilingual (EN/CN) capability
Backend	Python (FastAPI) + Celery	FastAPI for the REST API layer; Celery with Redis for async task queue handling parallel supplier outreach and scheduled follow-ups
Email	Microsoft Graph API or SMTP/IMAP	Graph API preferred if on Microsoft 365; otherwise standard SMTP/IMAP with a dedicated AI email account (e.g., purchasing-ai@aptech.com)
Database	PostgreSQL + pgvector	Relational DB for order tracking and supplier records; vector extension for semantic search over supplier capabilities and past conversations
Search	SerpAPI + custom CN crawlers	SerpAPI for English web search; custom integration with Baidu and 1688.com APIs for Chinese supplier sourcing
Frontend	React + TailwindCSS	Chat interface for engineers, dashboard for purchasing team oversight, and a quote comparison view
Logging	Markdown files in Git repo or S3	Each PO gets a structured .md file with timestamped entries; stored in version control for auditability

Table 2: *Recommended Technology Stack*

4.4 Key Capabilities Mapped to Requirements

Requirement	AI Capability	Target Phase
Parallel processing of POs from multiple engineers	Async task queue processes multiple requests concurrently	Phase 1 (MVP)
Exhaust all potential suppliers (lead time, price, quality)	Internal DB query + web search + CN internet sourcing	Phase 2
Negotiate on behalf of purchasing team	AI negotiation engine with pre-approved parameters	Phase 3
Central point of communication	AI Communication Hub with routing logic	Phase 1 (MVP)
Automated follow-ups and reminders	Scheduled task runner with escalation rules	Phase 2
Chinese internet supplier sourcing	Baidu/1688/Alibaba API integration + translation	Phase 2
Query internal AP Tech supplier database	Direct DB integration with approved vendor list	Phase 1 (MVP)
Operate a dedicated email account	SMTP/IMAP or Graph API email integration	Phase 1 (MVP)
Conversations retained in markdown format	Structured MD logging per PO with timestamps	Phase 1 (MVP)

Table 3: *Requirements-to-Capability Mapping*

5. Implementation Roadmap

5.1 Phase 1: MVP (Weeks 1–6)

Goal: Replace the manual email relay and establish the foundational AI infrastructure.

The MVP focuses on the highest-impact, lowest-risk capabilities that address the core bottleneck. By the end of Phase 1, engineers can submit purchase requests through a chat interface, the AI queries the internal supplier database, drafts outreach emails for purchasing team approval, and logs all conversations in markdown format.

MVP deliverables:

1. Chat interface for engineers to submit purchase requests with specifications
2. Integration with AP Tech's internal approved supplier database
3. AI-drafted email generation for supplier outreach (with human approval before sending)
4. Structured markdown conversation logging per purchase order
5. Basic purchase order status dashboard

5.2 Phase 2: Enhanced Capabilities (Weeks 7–12)

Goal: Enable parallel processing, Chinese supplier sourcing, and automated follow-ups.

Phase 2 removes the sequential bottleneck by enabling the AI to contact multiple suppliers simultaneously for the same request. It also adds Chinese internet sourcing capabilities and automated reminder scheduling.

Phase 2 deliverables:

1. Parallel multi-supplier outreach for each purchase request
2. Automated follow-up reminders when supplier/engineer responses are overdue
3. Chinese internet sourcing integration (1688.com, Alibaba, Baidu)
4. Quote comparison dashboard (price, lead time, quality side-by-side)
5. Template-based negotiation for standard scenarios

5.3 Phase 3: Intelligent Automation (Weeks 13–20)

Goal: Introduce AI-driven negotiation, predictive analytics, and full order lifecycle management.

Phase 3 deliverables:

1. AI-driven price and lead time negotiation within pre-approved parameters
2. Predictive lead time analysis based on historical data
3. Automated PO tracking with milestone alerts and delay predictions
4. Supplier performance scoring and recommendation engine

5. ERP/inventory system integration

5.4 Phase 4: Scale & Optimize (Weeks 21+)

Goal: Organization-wide rollout, analytics, and continuous improvement.

Phase 4 extends the system beyond the initial team, adds spend analytics and compliance automation, and establishes a continuous learning loop where the AI improves its supplier recommendations and negotiation strategies based on outcomes.

5.5 Target KPIs

Metric	Current State	Target (Phase 2)	Target (Phase 4)
Emails per purchase	30+	8–12	3–5
Cycle time (request to confirmation)	~2 weeks	5–7 days	2–3 days
Suppliers contacted per request	1–2 (sequential)	5–10 (parallel)	10–20+
Tracking error rate	High (manual)	Low (semi-auto)	Near zero (automated)
Purchasing team time on relay work	>60%	<20%	<5%

Table 4: *Target KPI Progression*

6. Risks & Mitigations

Risk	Severity	Mitigation	Phase
AI sends incorrect specs to supplier	High	Human-in-the-loop approval for all outbound emails in MVP; confidence scoring before auto-send in later phases	Phase 1: mandatory approval; Phase 3: conditional auto-send
Supplier resistance to AI-generated emails	Medium	Emails sent from a human-branded account; natural language tone; purchasing team CC'd for credibility	All phases
Chinese internet access restrictions or data quality	Medium	Use established APIs (1688 Open Platform); validate supplier legitimacy through cross-referencing before outreach	Phase 2
Internal DB integration complexity	Low–Med	Early technical spike in Week 1 to assess DB schema and access; fallback to CSV import if direct integration is delayed	Phase 1
Over-reliance on AI without oversight	Medium	Purchasing dashboard shows all AI actions; escalation rules ensure human review for high-value or unusual requests	All phases

Table 5: *Risk Assessment Matrix*

7. Conclusion & Next Steps

AP Tech's current purchase process suffers from structural inefficiencies rooted in manual relay communication, sequential processing, limited supplier reach, and error-prone tracking. The evidence — 30+ emails and a 2-week cycle for a single purchase — confirms that these are not minor inconveniences but systemic bottlenecks that scale poorly as the organization grows.

The proposed AI Purchasing Agent addresses each inefficiency directly. The phased implementation ensures that the team sees value quickly (Phase 1 MVP in 6 weeks) while building toward full autonomous purchasing capabilities. The MVP prioritizes the highest-impact changes: eliminating the email relay, enabling internal database queries, operating a dedicated email account, and producing markdown conversation logs.

Recommended immediate next steps:

6. Conduct a technical spike on the AP Tech supplier database to assess schema, access methods, and data quality (Week 1)
7. Set up the dedicated AI email account and configure SMTP/IMAP or Graph API access (Week 1)
8. Build and demonstrate the chat-to-email MVP prototype with a single end-to-end purchase flow (Weeks 2–4)
9. Gather feedback from 2–3 engineers and the purchasing team lead on the MVP (Weeks 5–6)
10. Refine and plan Phase 2 based on MVP learnings (Week 6)